

MAKE UP EXAMINATIONS – JUNE 2023

Program	: B.E :- Computer Science and Engineering	Semester	: III
Course Name	: Digital Design and Computer Organization	Max. Marks	: 100
Course Code	: CS32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Simplify the following function using K-Map, obtain the minimum SOP solution and realize the circuit using NAND gates only.
 $F(A,B,C,D) = \sum m(0, 1, 2, 3, 4, 5, 8, 9, 10, 14, 15)$. CO1 (06)
 - Implement the following function using a 4:1 Multiplexer using the appropriate method. $F(A,B,C) = \sum m(1,2,6,7)$. CO1 (06)
 - Write a VHDL code for 4:1 MUX. CO1 (08)
 - Design 2-to-4 Decoder with enable input E.
- Simplify the following function using K-Map, obtain the minimum POS solution and realize the circuit using NOR gates only.
 $F(A,B,C,D) = \prod M(0, 3, 4, 5, 8, 9, 10, 14, 15)$. CO1 (06)
 - Design BCD-to-7-segment decoder. CO1 (08)
 - Determine the Essential Prime Implicant using Tabular Method for the given function. $F(A,B) = \sum m(0,2,6,7) + d(3)$. CO1 (06)

UNIT - II

- Explain the working of JK flip flop with the help of circuit diagram, characteristic table and timing diagram. CO2 (06)
 - Design a mod 8 asynchronous UP counter using any FFs of your choice and draw the relevant waveforms. Briefly discuss its working. CO2 (08)
 - Derive the characteristic equation for D-flipflop and SR-flipflop. CO2 (06)
- Design Mod 6 synchronous down counter using T-FF. CO2 (08)
 - Design a sequence detector circuit that detects if two or more consecutive 1s occur on its input w. i.e. The output z is equal to 1 if during two immediately preceding clock cycles the input w was equal to 1. Otherwise, the value of z is equal to 0. CO2 (08)
 - Explain the working of SISO shift register with a timing diagram. CO2 (04)

UNIT - III

- Explain different functional units of computer with block diagram. Mention the functions of following registers i) MAR ii) PC. CO3 (08)
 - Represent the decimal values +5, -4 and -7 in the following binary formats:
i) Sign and Magnitude ii) 1's complement iii) 2's complement. CO3 (06)
 - Explain Big-endian and Little-endian with example. CO3 (06)
- Represent the following decimal values using 5bit binary numbers and subtract them. Check in each case whether overflow occurs.
i) +5 and + 8 ii) +6 and -12 CO3 (06)
 - Explain the basic performance equation. CO3 (04)

- c) Define addressing mode. Explain the following addressing modes with an example for each: CO3 (10)
 i) indirect addressing mode ii) indexed addressing mode
 iii) Relative addressing mode iv) auto decrement addressing mode.

UNIT- IV

7. a) Multiply +13 and -6 using bit-pair recording method. CO4 (05)
 b) Explain with example three bus organization of the data path. CO4 (10)
 c) Write a note on data hazards in pipelining. CO4 (05)
8. a) With a neat diagram, explain the floating point addition/subtraction unit. CO3 (10)
 b) Write and explain the control sequences for the execution of following instruction: Add (R3), R1. CO4 (05)
 c) Write a note on instruction hazards. CO4 (05)

UNIT - V

9. a) i) How many total bits are required for a direct-mapped cache with 16 KB of data and 4-word blocks, assuming a 32-bit address? CO5 (08)
 ii) Consider a cache with 64 blocks and a block size of 16 bytes. What block number does byte address 1200 map to?
 b) Explain I/O interface for an input device with a neat diagram. CO5 (06)
 c) Discuss process of Enabling and Disabling interrupts. CO5 (06)
10. a) Compare and contrast the cache mapping techniques with an example. CO5 (08)
 b) Discuss the transfer of control through the use of interrupts. CO5 (08)
 c) Explain the use of Cache memory with a neat diagram. CO5 (04)
